CSC/SDS 235: Visual Analytics

Final Project: Putting it all together for a Domain Expert

This assignment should be completed in groups of 3– working with a different size group (including solo) must be approved prior to submitting your first milestone. You may form groups with students in either section of the class.

Goals:

- Build a novel visual analytics system for a domain expert

Instructions

We spent the semester learning the building blocks of visual analytics: analysis approaches, visualization techniques, and coding skills. In this project you will put these pieces together to build a visual analytic system for a domain expert. Your project will be similar to a visualization design study: *a project in which visualization researchers analyze a specific real-world problem faced by domain experts, design a visualization system that supports solving this problem, validate the design, and reflect about lessons learned in order to refine visualization design guidelines* [1]. In the interest of time, we will loosely follow design study lite methodology [2].

Four domain experts have agreed to work with us. Here are synopses of their research areas:

- **Marney Pratt (Biology):** Marney has been collecting data monitoring the Mill River up- and downstream of paradise pond for several years. She would like a visual analytics system to help her analyze and communicate this data in an efficient way.
- **Mariana Abarca (Biology):** Mariana is working on a new project studying moths with data collected via images. The image data is translated into tabular form, and she would like a visual analytics system to help her explore and make sense of the data.
- **Sinead Keogh (Public Health):** Sinead works with the Public Health Institute of Western MA. The Public Health Institute is working on a public-facing dashboard that visualizes absenteeism across the 4 western MA counties. *The output of this project will be a Tableau Dashboard that is published for public use*.
- **Becca Malloy (CEEDS):** Becca is the assistant director of sustainability for CEEDS. She has a wealth of data on Smith's carbon emissions, attitudes towards sustainability on campus, and sustainability of the products purchased by Smith dining. She would like a visual analytics system that helps non-expert users explore CEEDS data.

We will need at least one group for each project, so that each domain expert ends up with at least one visual analytic system.

To help manage the workload of designing and implementing and visual analytic system, your project is broken into milestones that will be submitted weekly. Milestone details follow.

Milestone 1 (fp01): Project Plan [18pts]

This milestone focuses on the **discover** and **design** steps of the design study methodology.

A good starting point to this milestone is to read sections 4.2.1 and 4.2.2 of *Design Study Methodology: Reflections from the Trenches and the Stacks* [1].

Develop a semi-structured interview script to use for the abstract phase of your design study. Your script should include questions to ascertain information about your domain expert's data, their high-level analysis needs/questions, and the intended user of the VA system. Aim for an interview that will last roughly 45 minutes.

All groups working with the same domain expert must meet with Ab to review their interview scripts prior to interviewing the expert. **Coordinate a time to meet with Ab** to review and finalize your interview script. All group members must attend the meeting.

Once your script is finalized, **choose two delegates to interview your domain expert**. Schedule a time to interview your domain expert, remember to be respectful of their time. During the interview, **ask to record the interview or take thorough notes** to refer to after (you may not record someone without their explicit consent).

Finally, review the results of your interview. **Identify three specific high-level interactions** that your VA system will support. **Sketch a preliminary design** of your system and **identify at least one low-level interaction that supports each high-level interaction**. Your final VA system must include (1) at least three different visual encodings, (2) at least one details-on-demand interaction, (3) at least one coordinated view of the data.

Deliverables for this milestone:

1. Interview script
2. Meeting with Ab to finalize interview script
3. Interview of domain expert (evidenced by recording / notes)
4. Three high-level interactions
5. Preliminary system design
6. Three low-level interactions

Each deliverable will be scored from 0-3: 0 == Not completed, 1 == Needs improvement, 2 == Needs minor improvement, 3 == Fully completed.

Milestone 2 (fp02): Prototyping [9pts]

This milestone focuses on the **implement** and **deploy** steps of the design study methodology. A good starting point for this milestone is to read section 4.2.3 of *Design Study Methodology: Reflections from the Trenches and the Stacks* [1]. If you want more information about prototyping, a good starting point is *The prototyping in Interaction Design* [3].

Bring your sketch from milestone 1 to class the day fp01 is due. **Pair up with a team working with a different domain expert than you and provide feedback on each other's sketches**. Feedback should include: (a) Do you understand the other team's three high-level interactions? (b) Do you believe the other team's proposed design supports all three high-level interactions?

(c) Does the team's sketch include appropriate data-visual mappings and meet the final requirements listed in milestone 1? (d) What part(s) of the other team's design do you like most? (e) What suggestions do you have for improvement? **Record your feedback** so that you can hand it in as evidence that you completed this part of the milestone.

After receiving feedback, make any adjustments you deem necessary to your sketch and **begin implementing your system**.

Deliverables for this milestone:

1. Feedback on VA sketch
2. Revised VA sketch
3. In-process code base

Each deliverable will be scored from 0-3: 0 == Not completed, 1 == Needs improvement, 2 == Needs minor improvement, 3 == Fully completed.

Milestone 3 (fp03): End User Feedback [9pts]

This milestone focuses on the **implement** and **deploy** steps of the design study methodology.

Continue implementing your system until you have a minimally viable working version.

Coordinate a time to meet with your domain expert, plan on meeting with them for half an hour. Remember to plan in advance and be courteous of their time. During your meeting, show your domain expert your VA system and ask for feedback. Take notes on the feedback you receive from the domain expert.

After your meeting, review the feedback you received and identify what changes should be made to your VA system. **Identify two changes** in your list of changes that you can realistically complete within the remaining time for this project.

Continue implementing your system, including the changes identified from expert feedback.

Deliverables for this milestone:

1. Notes on feedback from domain expert
2. List of changes based on feedback, with two to be completed identified
3. In-process code base

Each deliverable will be scored from 0-3: 0 == Not completed, 1 == Needs improvement, 2 == Needs minor improvement, 3 == Fully completed.

Milestone 4 (fp04): Final Product [24pts]

This milestone focuses on the **disseminate** step of the design study methodology.

Finish implementing your VA system. By the due date of this milestone, it should be in a place where you can give it to your domain expert for them to use. Be sure your final product meets the requirements laid out in milestone 1.

Write up a README to include in your repository that details (a) the purpose of the VA system, (b) how to use the system, (c) any known bugs, and (d) documents any data cleaning/analysis.

Deliverables for this milestone:

1. Final VA system with:
    a. at least three different visual encodings [3pts]
    b. at least one details-on-demand interaction [3pts]
    c. at least one coordinated view of the data [3pts]
    d. appropriate data-visual mappings, titles, labels, etc. [3pts]
    e. reliable interactions [3pts]
    f. no console errors [3pts]
    g. modular code split appropriately between HTML/CSS/JS [3pts]
2. README [3pt]

Submission

Submit each milestone on Gradescope. How to submit as a group: https://guides.gradescope.com/hc/en-us/articles/21863861823373-Adding-Group-Members-to-a-Submission.

Works Cited

[1] M. Sedlmair, M. Meyer and T. Munzner, "Design Study Methodology: Reflections from the Trenches and the Stacks," IEEE Transactions on Visualization and Computer Graphics, *pp. 2431-2440, Dec 2012.*

[2] U. Haque Syeda, P. Murali, L. Roe, B. Berkey and M. Borkin, "Design Study "Lite" Methodology: Expediting Design Studies and Enabling the Synergy of Visualization Pedagogy and Social Good," in In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*, New York, NY, 2020.*

[3] M. Hua and H. Qiu, "The Prototype in Interaction Design," in 9th International Conference on Computer-Aided Industrial Design and Conceptual Design*, Beijing, 2008.*